Hard calibration

Hard calibration applies a list of named parameter changes from `calibration.cal`, re-initializes all model objects, then re-runs the time loop. Each change names a parameter, a change type (`absval`, `abschg`, `pctchg`), a value, a list of objects (HRUs, channels, aquifers, plants, …) to apply the change to, and optional conditions. The parameter universe and the absolute min/max for each parameter are defined in `cal_parms.cal`.

Source

- Driver: src/calhard_control.f90
- Parameter selector: src/cal_parm_select.f90
- Change application with conditions: src/cal_conditions.f90
- Parameter database reader: src/cal_parm_read.f90
- Change list reader: src/cal_parmchg_read.f90

Workflow

1. The model is run once normally. Output baseline is captured.
2. The user edits `calibration.cal` with the parameter changes to test (or an external driver writes it).
3. SWAT+ is re-launched. During startup, `cal_parm_read` loads `cal_parms.cal` into the `cal_parms` array and `cal_parmchg_read` loads `calibration.cal` into the `cal_upd` array. Each `cal_upd` entry is crosswalked to its `cal_parms` entry by name to pick up the absolute bounds and object type.
4. Inside the simulation, `cal_conditions` is called once. For every entry in `cal_upd`, it evaluates the conditions, then walks the element list and calls `cal_parm_select` for each affected object. `cal_parm_select` does the actual write into the live model state (HRU, channel, aquifer, plant, soil layer, climate series).
5. The master switch `cal_hard = "y"` triggers a second pass: `calhard_control` calls `re_initialize` then `time_control`, re-running the full simulation with the modified parameters.

The change type drives how the value is applied to the existing parameter:

`chg_typ`	Effect
`absval`	The new value replaces the existing one.
`abschg`	The value is added to the existing one (signed).
`pctchg`	The value is interpreted as a percent change.

After every change the result is clamped to the `[absmin, absmax]` range from `cal_parms.cal`.

Object types

`cal_parms.cal` tags each parameter with an object type (`ob_typ`). The type determines which collection in the model state is iterated when the change is applied:

Code	Collection	Examples
`hru`	HRUs	`cn2`, `ovn`, `lat_ttime`, `canmx`
`hlt`	`hru_lte`	parameters of the simplified HRU
`sol`	Soil layers within an HRU (`lyr1`/`lyr2` select the range)	`awc`, `bd`, `k`
`lyr`	Soil layers	layer-targeted variants
`plt`	Plants	`lai_pot`, `harv_idx`, `epco`, `phu_mat`

Code	Collection	Examples
`aqu`	Aquifers	`alpha`, `revap_co`, `flo_min`
`cha`, `sdc`, `rte`	Channels (legacy, degraded, routed)	channel routing and sediment parameters
`swq`	Channel water quality	nutrient parameters
`res`	Reservoirs	`res_pvol`, surface area parameters
`rdt`	Reservoir decision tables	`drawdown_days`, `withdraw_rate`
`ru`	Routing units	RU-level parameters
`bsn`	Basin	basin-wide scalars in `parameters.bsn`
`pcp`, `tmp`, `cli`	Climate gauges and series	`precip`, `temp` adjustments over date ranges
`gwf`	gwflow cells	gwflow parameters

The list above is taken from the `select case (cal_parms(ipar)%ob_typ)` block in `cal_parmchg_read.f90` and the dispatch in `cal_conditions.f90`.

Re-initialization

`calhard_control.f90` calls `re_initialize` before re-running. This resets simulation state (water content, plant state, accumulators) so the second run is comparable to the baseline rather than continuing from baseline end-state.

Related pages

- Parameter change files. Column-level format of `cal_parms.cal` and `calibration.cal`.
- Conditions and regionalization. Limiting a change to HRUs that match a condition.
- Sensitivity workflow. Driving hard calibration from an external sampler.
- Soft calibration. Adjust process outputs toward regional targets before fitting to time series.